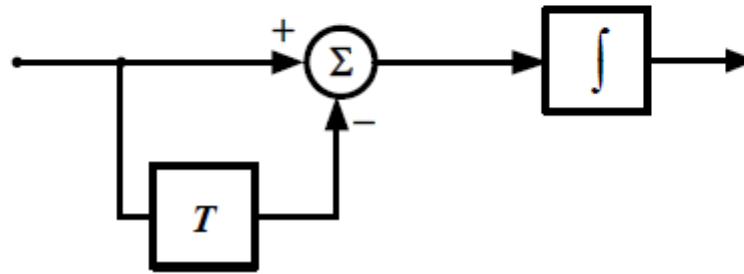
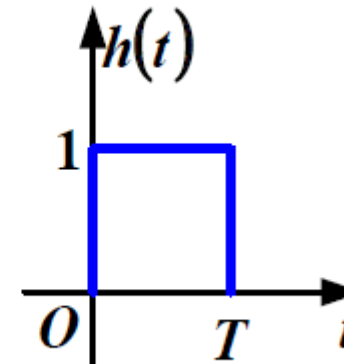


Example 1

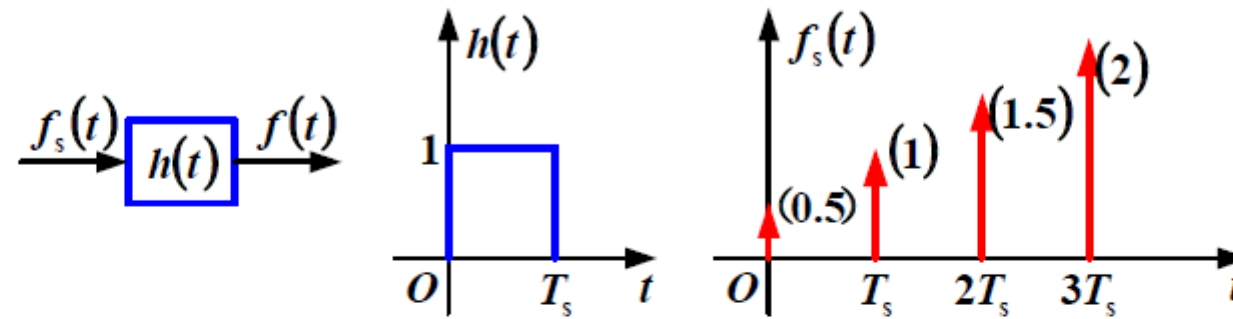
Please find the unit impulse response of the system shown in the figure



$$\begin{aligned} h(t) &= \int_{-\infty}^t [\delta(\tau) - \delta(\tau - T)] d\tau \\ &= u(t) - u(t - T) \end{aligned}$$

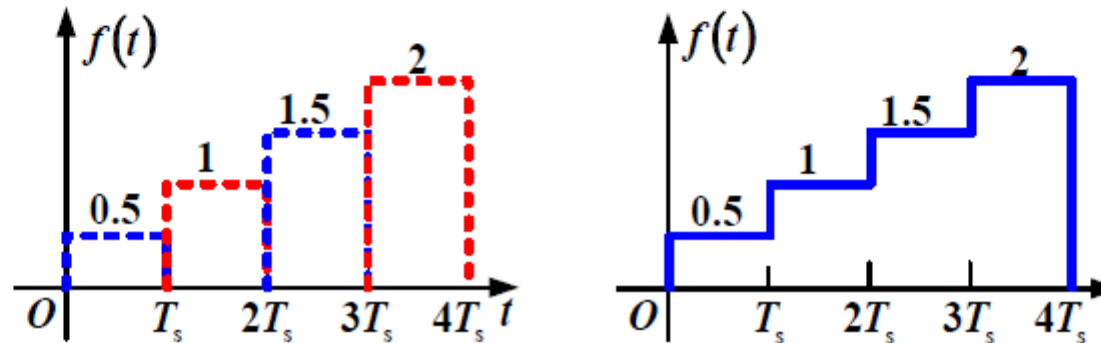


Example 2



Please draw the waveform of the signal $f(t)$

$$f(t) = f_s(t) * h(t)$$



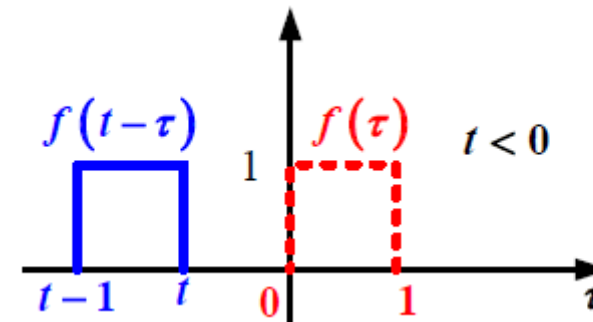
Example 3

$f(t) = u(t) - u(t-1)$, find convolution $s(t) = f(t) * f(t)$

(1) Graphical method Floating coordinate

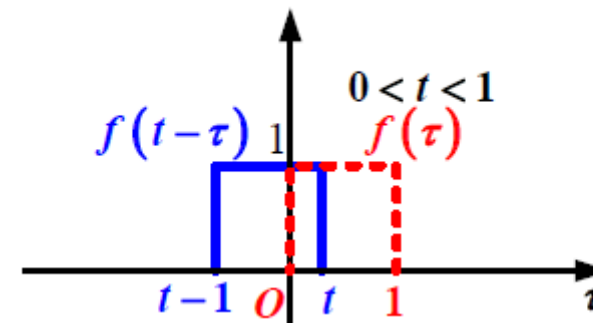
(a) When $t < 0$,

$$s(t) = 0$$



(b) When $0 < t < 1$,

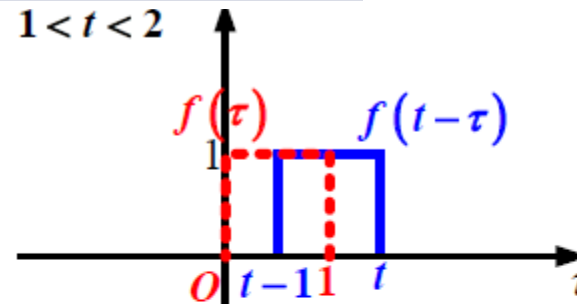
$$s(t) = \int_0^t 1 d\tau = t$$



Example 3

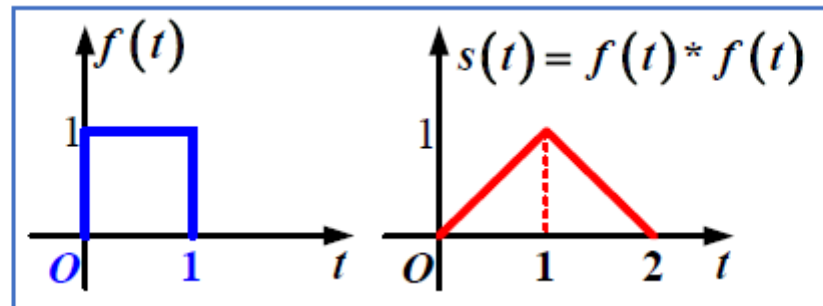
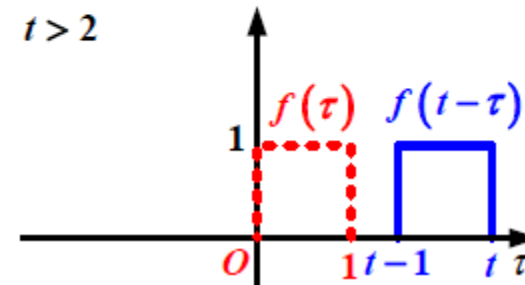
(c) When $1 < t < 2$,

$$s(t) = \int_{t-1}^1 1 d\tau = 2 - t$$



(d) When $t > 2$,

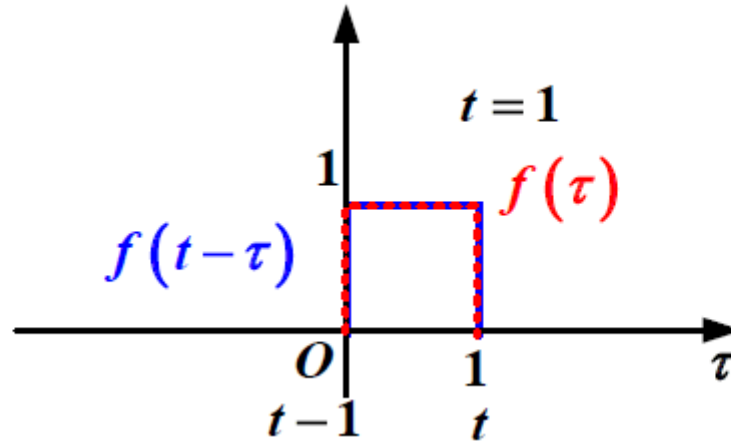
$$s(t) = 0$$



Example 3

(2) When does the output reach its maximum value?

When two pulses overlap, the integration result is the highest, $t=1$



Example 4

The unit impulse response of a system is $h(t) = e^{-2t}u(t)$,

Determine the causality and stability of the system

(1) Causality

$$\because t < 0, \quad h(t) = 0$$

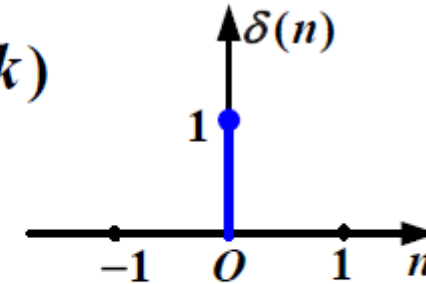
(2) Stability

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-\infty}^{\infty} e^{-2t} u(t) dt = \frac{1}{-2} e^{-2t} \Big|_0^{\infty} = \frac{1}{2} < \infty$$

Example 5

The accumulator can be represented as $y(n) = \sum_{k=-\infty}^n x(k)$

Determine the causality and stability of the system



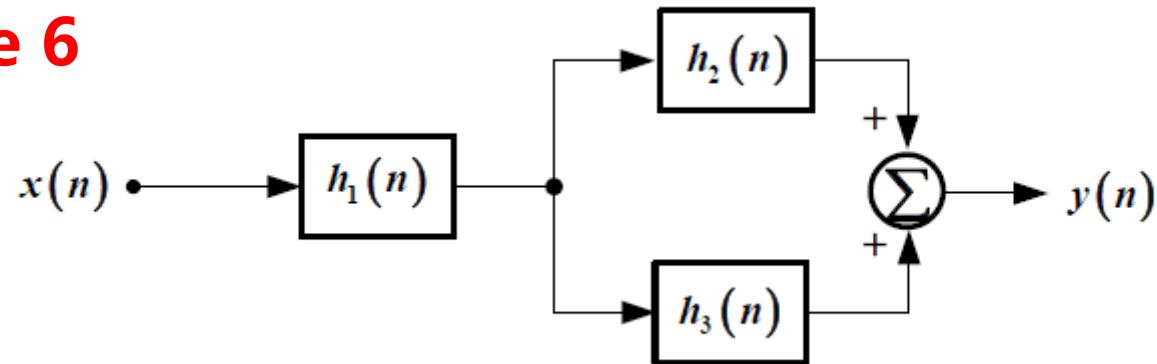
The unit impulse/sample response of the system is

$$h(n) = \sum_{k=-\infty}^n \delta(k) = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} = u(n)$$

$$n < 0, h(n) = 0 \quad \therefore \text{Causal}$$

$$\sum_{n=-\infty}^{\infty} |h(n)| = \infty \quad \therefore \text{Not stable}$$

Example 6



An interconnected linear time-invariant (LTI) system is shown in the figure above. The unit sample responses of its subsystems are

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n+2), \quad h_2(n) = \delta(n), \quad h_3(n) = u(n-1)$$

The unit impulse response of the system?

$$\begin{aligned}h(n) &= h_1(n) * [h_2(n) + h_3(n)] \\&= \left(\frac{1}{2}\right)^n u(n+2) * [\delta(n) + u(n-1)] \\&= \left(\frac{1}{2}\right)^n u(n+2) * u(n) \\&= \sum_{m=-\infty}^{\infty} \left(\frac{1}{2}\right)^m u(m+2) u(n-m)\end{aligned}$$